

Polynomials degree stress prototype: Construction

Find degree after like terms cancel

Subject: Mathematics

Topic: Polynomials

Concept: Find degree after like terms cancel

Concept difficulty: Medium concept

You need

- Can identify a term's exponent and combine like terms.

The key step

Degree belongs to the simplified polynomial, so a written high-power term whose total coefficient is zero does not decide it.

Simplify before naming degree.

1. **Combine terms with the same exponent.**

Why valid: Only like powers share a coefficient.

Result: $3x^4 - 3x^3 + 2x + 5$ becomes $2x + 5$.

2. **Inspect the largest exponent still attached to a nonzero coefficient.**

Why valid: This is the degree rule on NCERT page 18.

Result: The degree is 1, not 3.

power 0

power 1

power 2

power 3

Stage 1 of 3

Show powers 0, 1, 2 and 3 as coefficient slots before selecting the highest nonzero one.

Worked example

Find the degree of $5x^2 + 3x - 5x^2 + 4$.

1. The x^2 coefficients add to zero.

2. The result is $3x + 4$, whose highest nonzero power is 1.

Degree 1.

A tempting wrong path

- The largest exponent printed anywhere is always the degree.

Diagnose

- In $3x^3 - 3x^3 + 2x + 5$, what remains of the x^3 coefficient?

Repair

- Combine the x^3 terms first, then inspect the powers that remain.

Check it independently

- Expand and combine again; verify every higher exponent has coefficient zero.

Exit task

What is the degree of $4x^2 - 4x^2 + 7$? Explain the cancellation.

Model answer

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